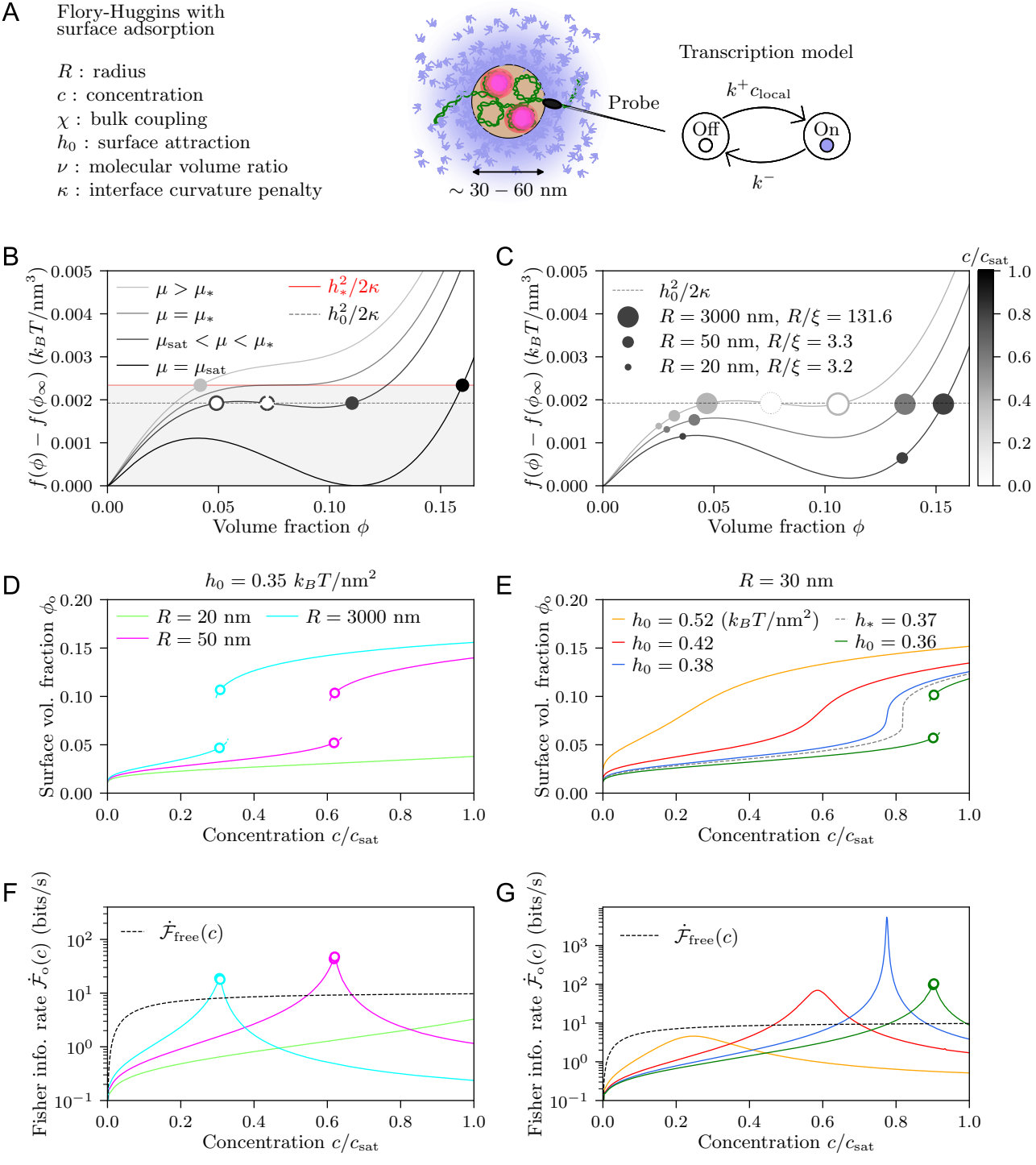

A
Flory-Huggins with surface adsorption
$R$ : radius
$c$ : concentration
$\chi$ : bulk coupling
$h_0$ : surface attraction
$\nu$ : molecular volume ratio
$\kappa$ : interface curvature penalty
Probe
$\sim 30-60$ nm
Transcription model
$k^+ c_{\text{local}}$
Off
On
$k^-$
B
$f(\phi) - f(\phi_\infty)$ $(k_BT/\text{nm}^3)$
$\mu > \mu_*$
$\mu = \mu_*$
$\mu_{\text{sat}} < \mu < \mu_*$
$\mu = \mu_{\text{sat}}$
$h_*^2/2\kappa$
$h_0^2/2\kappa$
Volume fraction $\phi$
C
$c/c_{\text{sat}}$
$h_0^2/2\kappa$
$R = 3000$ nm, $R/\xi = 131.6$
$R = 50$ nm, $R/\xi = 3.3$
$R = 20$ nm, $R/\xi = 3.2$
Volume fraction $\phi$
D
$h_0 = 0.35\ k_BT/\text{nm}^2$
Surface vol. fraction $\phi_o$
$R = 20$ nm
$R = 3000$ nm
$R = 50$ nm
Concentration $c/c_{\text{sat}}$
E
$R = 30$ nm
Surface vol. fraction $\phi_o$
$h_0 = 0.52$ $(k_BT/\text{nm}^2)$
$h_0 = 0.42$
$h_0 = 0.38$
$h_* = 0.37$
$h_0 = 0.36$
Concentration $c/c_{\text{sat}}$
F
Fisher info. rate $\dot{\mathcal{F}}_o(c)$ (bits/s)
$\dot{\mathcal{F}}_{\text{free}}(c)$
Concentration $c/c_{\text{sat}}$
G
Fisher info. rate $\dot{\mathcal{F}}_o(c)$ (bits/s)
$\dot{\mathcal{F}}_{\text{free}}(c)$
Concentration $c/c_{\text{sat}}$